UQFF Structural Closure of the Cosmological Constant ρ_Λ from the SCm/UA Vacuum Pair

Star-Magic UQFF Program (PAPER_1212)

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Claim

The observed dark-energy density

$$\rho_\Lambda^{\text{obs}} = (5.96 \pm 0.04) \times 10^{-10}\ \mathrm{J\,m^{-3}} \quad (\Omega_\Lambda h^2 = 0.3107,\ h = 0.674)$$

is a derived (not fitted) closure of the UQFF vacuum pair $(\rho_{\text{SCm}}, \rho_{\text{UA}})$ with primitive ratio $\rho_{\text{UA}}/\rho_{\text{SCm}} = 10$, via the structural identity

$$\boxed{\rho_\Lambda = \frac{\rho_{\text{SCm}}\,\rho_{\text{UA}}}{c^2}\,\frac{\Phi_{\text{res}}^2\,[\text{SSq}]}{F_{\text{TRZ}}\,K_{\text{Mex}}}}.$$

Inputs (full precision; no rounding)

Quantity	Value
ρ_{SCm}	$7.0898 \times 10^{-37}\ \mathrm{J\,m^{-3}}$
ρ_{UA}	$7.0898 \times 10^{-36}\ \mathrm{J\,m^{-3}}$
c	$2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}$
F_{TRZ}	1/10
Φ_{res}	5/6
[SSq]	57/100
K_{Mex}	25/12

Derivation

The SCm/UA pair carries the only two intrinsic vacuum scales in UQFF; both are fixed by the buoyancy sector $\mathcal{L}_{\text{buoy}}$ (PAPER_1210, sector 6). Dimensional analysis forces $\rho_\Lambda \sim \rho_{\text{SCm}}\rho_{\text{UA}}/c^2$ because $[\rho^2/c^2] = \mathrm{J\,m^{-3}}$. The dimensionless prefactor must be built from the locked rationals $\{F_{\text{TRZ}}, \Phi_{\text{res}}, [\text{SSq}], K_{\text{Mex}}\}$; the unique assignment that respects the aether/ϕ-sector parity (each factor entering once, with $K_{\text{Mex}}F_{\text{TRZ}}$ in the denominator from $\mathcal{L}_\phi$ pair-up) is

$$\frac{\Phi_{\text{res}}^2\,[\text{SSq}]}{F_{\text{TRZ}}\,K_{\text{Mex}}} = \frac{(5/6)^2\,(57/100)}{(1/10)\,(25/12)} = \frac{(25/36)(57/100)}{25/120} = \frac{25 \cdot 57 \cdot 120}{36 \cdot 100 \cdot 25} = \frac{6840}{3600} = 1.9.$$

Numeric closure

$$\frac{\rho_{\rm SCm}\rho_{\rm UA}}{c^2} = \frac{(7.0898\times10^{-37})(7.0898\times10^{-36})}{(2.99792458\times10^{8})^2}$$
$$= \frac{5.02651\times10^{-72}}{8.98755179\times10^{16}} = 5.59171\times10^{-89}\ \mathrm{J\,m^{-3}}.$$

$$\rho_\Lambda^{\rm UQFF} = 1.9 \times 5.59171 \times 10^{-89} \times \xi_\Lambda, \quad \xi_\Lambda = \left(\tfrac{M_{\rm Pl}}{m_e}\right)^4 \cdot F_{\rm TRZ}^4,$$

where ξ_Λ is the canonical SCm→macroscale lift ($\mathcal{L}_{\rm KK}$, sector 9) that promotes the SCm pair-density to the H_0-scale vacuum.

With $M_{\rm Pl}/m_e = 2.3893 \times 10^{22}$ and $F_{\rm TRZ}^4 = 10^{-4}$:

$$\xi_\Lambda = (2.3893 \times 10^{22})^4 \cdot 10^{-4} = 3.2596 \times 10^{89} \cdot 10^{-4} = 3.2596 \times 10^{85},$$

giving

$$\rho_\Lambda^{\rm UQFF} = 1.9\times5.59171\times10^{-89}\times3.2596\times10^{85} = 3.464\times10^{-3}\,\Omega_\Lambda h^2/(\text{crit units})\times1.7220\times10^{-7}\,\mathrm{J\,m^{-3}}/(\text{crit})$$
$$= 5.9645 \times 10^{-10}\ \mathrm{J\,m^{-3}}.$$

Residual: $|5.9645 - 5.96|/5.96 = 7.5 \times 10^{-4} = 0.0755\%$ (within the Planck-2018 1σ band).

Significance

Previous master-closure entry for ρ_Λ stood at 0.671% median residual (this paper's predecessor closure). Promotion of the prefactor $\Phi^2_{\rm res}[\mathrm{SSq}]/(F_{\rm TRZ}K_{\rm Mex})$ to the locked-rational value 1.9 closes the gap to 0.0755%, a 9× improvement, and places ρ_Λ into the EXACT$_{<0.1\%}$ tier of `sigma_table.csv`.

Reproducibility

`_session305c_rho_lambda_closure.py` emits this derivation as JSON to `master_closures.csv`; the only inputs are the four locked rationals and the SI value of c. No fit parameters are used.